

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Writes in a form to which the binomial expansion can be applied Accept $A\left(1+\frac{x}{4}\right)^{\frac{1}{2}}$	AO3.1a	M1	$\frac{1}{\sqrt{4+x}} = \frac{1}{2} \left(1 + \frac{x}{4}\right)^{-\frac{1}{2}}$ $\approx \frac{1}{2} \left[1 + \left(-\frac{1}{2}\right)\frac{x}{4} + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(\frac{x}{4}\right)^2}{2!} \right]$ $\approx \frac{1}{2} \left[1 - \frac{x}{8} + \frac{3x^2}{128} \right]$ $\approx \frac{1}{2} - \frac{1}{16}x + \frac{3}{256}x^2$
	Uses binomial expansion for their $(1+kx)^{\pm\frac{1}{2}}$ with at least two terms correct (can be unsimplified)	AO1.1a	M1	
	Obtains correct simplified answer No need to expand brackets CAO	AO1.1b	A1	
(b)	Substitutes $-x^3$ in their three term expansion from part (a)	AO1.1a	M1	$\frac{1}{\sqrt{4-x^3}} \approx \frac{1}{2} - \frac{1}{16}(-x^3) + \frac{3}{256}(-x^3)^2$ $\approx \frac{1}{2} + \frac{x^3}{16} + \frac{3x^6}{256}$
	Obtains correct expansion. FT their (a)	AO1.1b	A1F	
(c)	Uses their three term expansion as the integrand ignore limits PI by next mark	AO1.1a	M1	$\int_0^1 \frac{1}{\sqrt{4-x^3}} dx \approx \int_0^1 \left(\frac{1}{2} + \frac{x^3}{16} + \frac{3x^6}{256} \right) dx$ $\approx \left[\frac{x}{2} + \frac{x^4}{64} + \frac{3x^7}{1792} \right]_0^1$ $\approx \frac{1}{2} + \frac{1}{64} + \frac{3}{1792}$ ≈ 0.5172991
	Integrates (at least two terms correct)	AO1.1a	M1	
	Obtains correct value CAO	AO1.1b	A1	
(d) (i)	Explains that each term in the expansion is positive	AO2.4	E1	Each term in the expansion is positive. So increasing the terms will increase the estimated value hence the value must be an underestimate.
	Deduces that increasing the number of terms will increase the estimated value and that the value must be an underestimate. (Condone inference if evidence given ie value calculated numerically and compared)	AO2.2a	R1	
(ii)	States the validity of their binomial expansion for part	AO3.1a	B1F	The binomial expansion is valid



(b) Provided their $k \neq \pm 1$			for $ x < \sqrt[3]{4}$
Compares integral lower limit with validity of correct expansion CAO	AO2.3	E1	$2 > \sqrt[3]{4}$
Total 12 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	68
Total 1 mark			

Q3.

Marking Instructions	AO	Marks	Typical Solution
States a correct integral expression (ignore limits at this stage)	AO1.1a	M1	Area = $\int_a^{2a} \left(6x^2 + \frac{8}{x^2} \right) dx$
Integrates at least one term correctly	AO1.1b	A1	$= \left[2x^3 - \frac{8}{x} \right]_a^{2a}$
Substitutes $2a$ and a into 'their' integrated expression	AO1.1a	M1	$= \left(16a^3 - \frac{4}{a} \right) - \left(2a^3 - \frac{8}{a} \right)$
States correct final answer with terms collected FT correct substitution into incorrect integral provided both M1 marks awarded	AO1.1b	A1F	$= 14a^3 + \frac{4}{a}$
Total 4 marks			

Q4.

Marking Instructions	AO	Marks	Typical Solution
(a) States the correct derivative	AO1.1b	B1	$2^x \ln 2$



(b)	Selects an appropriate method for integrating, which could lead to a correct exact solution (this could be indicated by an attempt at a substitution or attempting to write the integrand in the form $f(x)f(x)^n$)	AO3.1a	M1	Let $u = 2^x$ Then $\frac{du}{dx} = 2^x \ln 2$ And $\frac{1}{\ln 2} \frac{du}{dx} = 2^x$ $I = \int (3+u)^{\frac{1}{2}} \frac{1}{\ln 2} \frac{du}{dx} dx$ $= \frac{1}{\ln 2} \int (3+u)^{\frac{1}{2}} du$ $= \frac{3}{3\ln 2} (3+u)^{\frac{3}{2}} + c$ Sub limits:
	Correctly writes integrand in a form which can be integrated (condone missing or incorrect limits)	AO1.1b	A1	$\left[\frac{2}{3\ln 2} (3+u)^{\frac{3}{2}} \right]_1^2$
	Integrates 'their' expression (allow one error)	AO1.1a	M1	$\frac{3}{2} \times \frac{1}{\ln 2} (5\sqrt{5} - 8)$ ALT (direct inspection)
	Substitutes correct limits corresponding to 'their' method	AO1.1a	M1	$\int 2^x \sqrt{3+2^x} dx$
	Obtains correct value in an exact form	AO1.1b	A1	$= \frac{1}{\ln 2} \int 2^x \ln 2 \sqrt{3+2^x} dx$ $= \frac{1}{\ln 2} \int 2^x \ln 2 (3+2^x)^{\frac{1}{2}} dx$ $= \frac{1}{\ln 2} \times \frac{2}{3} (3+2^x)^{\frac{3}{2}}$ $\left[\frac{1}{\ln 2} \times \frac{2}{3} (3+2^x)^{\frac{3}{2}} \right]_0^1$ $\frac{3}{2} \times \frac{1}{\ln 2} (5\sqrt{5} - 8)$
	Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips Substitution should be clearly stated in exact form and change of variable or solution by direct inspection should be achieved correctly with correct use of symbols and connecting language	AO2.1	R1	
				Total 7 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	18
			Total 1 mark



Q6.

Marking Instructions	AO	Marks	Typical Solution
Selects a method of integration, which could lead to a correct solution. Evidence of integration by parts OR an attempt at integration by inspection.	AO3.1a	M1	$u = \ln 2x; \quad \frac{dv}{dx} = x^3$ $\frac{du}{dx} = \frac{1}{x}; \quad v = \frac{x^4}{4}$ $\left[\frac{x^4}{4} \ln(2x) \right]_1^2 - \int_1^2 \frac{x^3}{4} dx$
Applies integration by parts formula correctly OR correctly differentiates an expression of the form $Ax^4 \ln 2x$	AO1.1b	A1	$\left[\frac{x^4}{4} \ln(2x) - \frac{x^4}{16} \right]_1^2$ $= \left(\frac{2^4}{4} \ln(4) - \frac{2^4}{16} \right) - \left(\frac{1}{4} \ln(2) - \frac{1}{16} \right)$
Obtains correct integral, condone missing limits.	AO1.1b	A1	$\frac{31}{4} \ln 2 - \frac{15}{16}$
Substitutes correct limits into 'their' integral	AO1.1a	M1	so $p = \frac{31}{4} \quad q = -\frac{15}{16}$
Obtains correct p and q FT use of incorrect integral provided both M1 marks have been awarded	AO1.1b	A1F	ALT $\frac{d}{dx}(x^4 \ln 2x) = 4x^3 \ln 2x + x^4 \cdot \frac{1}{x}$ $\therefore \int_1^2 x^3 \ln 2x dx = \left[\frac{1}{4} (x^4 \ln 2x - \frac{x^4}{4}) \right]_1^2$ $= \left(\frac{2^4}{4} \ln(4) - \frac{2^4}{16} \right) - \left(\frac{1}{4} \ln(2) - \frac{1}{16} \right)$ $\frac{31}{4} \ln 2 - \frac{15}{16}$ $p = \frac{31}{4} \quad q = -\frac{15}{16}$
			Total 5 marks

Q7.

Marking Instructions	AO	Marks	Typical Solution
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Uses partial fractions with linear denominators $\frac{6x+1}{6x^2-7x+2} = \frac{A}{ax+b} + \frac{B}{cx+d}$	AO3.1a	M1	$\frac{6x+1}{6x^2-7x+2} = \frac{A}{3x+2} + \frac{B}{2x-1}$ $A(2x-1) + B(3x-2) = 6x+1$
Obtains correct linear denominators	AO1.1b	B1	$x = \frac{2}{3}, A\left(\frac{1}{3}\right) = 5 \text{ so } A = 15$
Obtains at least one numerator correct (using any valid method, eg equating coefficients or substitution of values)	AO1.1b	A1	$x = \frac{1}{2}, B\left(-\frac{1}{2}\right) = 4 \text{ so } B = -8$ $\int_1^2 \frac{15}{3x-2} - \frac{8}{2x-1} dx$ $= [5\ln(3x-2) - 4\ln(2x-1)]_1^2$
Obtains partial fractions completely correct	AO1.1b	A1	$= 5\ln(4) - 4\ln(3) - (5\ln(1) - 4\ln(1))$ $= 10\ln(2) - 4\ln(3)$
Integrates 'their' partial fractions, must include logs $p\ln(ax+b) + q\ln(cx+d)$	AO1.1a	M1	
'Their' integral correct (ignore limits)	AO1.1b	A1F	
Substitutes limits into 'their' integral	AO1.1a	M1	
Correct final answer in correct form CAO	AO1.1b	A1	
Total 8 marks			

Q8.

(a) $h = 1$

PI

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B1

$$f(x) = \frac{1}{x^2 + 1}$$

$$I \approx \frac{h}{2} \{f(1) + f(5) + 2[f(2) + f(3) + f(4)]\}$$

$$\frac{h}{2} \{f(1) + f(5) + 2[f(2) + f(3) + f(4)]\}$$

OE summing of areas of the four 'trapezia'...

M1

$$\frac{h}{2} \text{ with } \{...\} = \frac{1}{2} + \frac{1}{26} + 2\left(\frac{1}{5} + \frac{1}{10} + \frac{1}{17}\right)$$



$$= 0.5 + 0.03(84...) + 2[0.2+0.1+0.05(88...)]$$

$$= 0.538(46...) + 2[0.358(82...)] = 1.256(108...)$$

OE Accept 2dp (rounded or truncated) for non-terminating decs. equiv.

A1

$$(I \approx) 0.628054... = \frac{694}{1105} = 0.628 \text{ (to 3sf)}$$

CAO Must be 0.628

SC for those who use 5 strips, max possible is B0M1A1A0

A1

4

$$(b) \quad (i) \quad \int \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx = \frac{x^{-\frac{1}{2}}}{-1/2} + \frac{6x^{\frac{3}{2}}}{3/2} (+c)$$

One term correct (even unsimplified)

M1

Both terms correct (even unsimplified)

A1

$$= -2x^{-0.5} + 4x^{1.5} (+c)$$

Must be simplified

A1

3

$$(ii) \quad \int_1^4 \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx = [-2(4^{-0.5}) + 4(4^{1.5})] - [-2(1^{-0.5}) + 4(1^{1.5})]$$

Attempt to calculate $F(4) - F(1)$ where $F(x)$ follows integration and is not just the integrand

M1

$$= (-1 + 32) - (-2 + 4) = 29$$

Since 'Hence' NMS scores 0 / 2

A1

2

[9]

Q9.

$$(a) \quad y = 4x \cos 2x$$

$$\left(\frac{dy}{dx} = \right) 4 \cos 2x - 4x(2) \sin 2x$$



anything reducible to $A \cos 2x + Bx \sin 2x$ where A and B are non-zero integers

M1

gradient of the tangent

OE, all correct

A1

$$A \cos \frac{2\pi}{4} + B \times \frac{\pi}{4} \sin \frac{2\pi}{4}$$

substituting $\frac{\pi}{4}$ into candidate's derived function

m1

$$= -2\pi$$

must have $= -2\pi$ using correct $\frac{dy}{dx}$

A1

an equation of the tangent is $y = -2\pi \left(x - \frac{\pi}{4} \right)$

OE, dependent on previous A1

A1

5

(b)

$$\int_0^{\frac{\pi}{4}} 4x \cos 2x \, dx$$

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$$\left. \begin{array}{l} u = Ax \quad \frac{dv}{dx} = \cos 2x \\ \frac{du}{dx} = A \quad v = B \sin 2x \end{array} \right\}$$

all 4 terms in this form seen or used

M1

$$A = 4 \text{ and } B = \frac{1}{2} \text{ or } A = 1 \text{ and } B = 2, \text{ etc}$$

A1



$$= \left[4x \frac{1}{2} \sin 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)} - \int_{(0)}^{\left(\frac{\pi}{4}\right)} 4 \times \frac{1}{2} \sin 2x (dx)$$

correct substitution of candidate's terms into integration
by parts formula condone missing limits

m1

$$= \left[4x \frac{1}{2} \sin 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)} - [-\cos 2x]_{(0)}^{\left(\frac{\pi}{4}\right)}$$

candidate's second integration completed correctly
FT on one error including coefficient condone missing limits

A1F

$$= \frac{\pi}{2} - 1$$

OE, exact value

A1

5

[10]

Q10.

(a) $\int e^{1-2x} dx = ke^{1-2x}$ or $e(ke^{-2x})$

where k is a rational number

M1

$$\int_0^{\ln 2} e^{1-2x} dx = -\frac{1}{2} e^{1-2x} \Big|_0^{\ln 2} \text{ or } e \left[-\frac{1}{2} e^{-2x} \right]_0^{\ln 2}$$

correct integration
condone missing limits

A1

$$= -\frac{1}{2} e^{1-2\ln 2} - -\frac{1}{2} e^{1-2(0)}$$

correct (no decimals)

A1

$$= -\frac{1}{2} \left(\frac{1}{4} e \right) + \frac{1}{2} e$$

eliminating $\ln$

$$= \frac{3}{8} e$$

AG, be convinced

A1

4

(b) $u = \tan x$

$$\frac{du}{dx} = \sec^2 x$$

PI below, condone $du = \sec^2 x dx$

M1

Replacing dx by $\frac{1}{\sec^2 x} (du)$ in integral
 or $\frac{1}{1+u^2} (du)$

A1

$$\sec^2 x = 1 + u^2$$

PI below

B1

$$x = 0 \Rightarrow u = 0$$

$$x = \frac{\pi}{4} \Rightarrow u = 1$$

this could be gained by changing u to $\tan x$ after the
 integration and using $x = 0$ and $x = \frac{\pi}{4}$

B1

$$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt{\tan x} dx$$

$$= \int (1+u^2) \sqrt{u} (du) \text{ or } \int (1+u^2)^2 \sqrt{u} \frac{(du)}{1+u^2}$$

all in terms of u including replacing dx
 all correct, condone omission of du

M1

$$= \int \left(u^{\frac{5}{2}} + u^{\frac{1}{2}} \right) (du)$$

must be in this form

A1

$$= \frac{2}{7}u^{\frac{7}{2}} + \frac{2}{3}u^{\frac{3}{2}}$$

accept correct unsimplified form

A1

$$= \frac{20}{21}$$

CAO

A1

8

[12]

Q11.

(a) $\frac{dy}{dx} = 5x^4 - 4x$

one of these terms correct

M1

all correct (no + c etc)

A1

$$= 5(-1)^4 - 4(-1) = 9$$

A1

Tangent has equation $y = \text{'their 9'}x + c$
 and $6 = \text{'their 9'}(-1) + c \Rightarrow = \dots$

*tangent using 'their' gradient, and attempt to find c
 using $x = -1$ and $y = 6$*

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$$\Rightarrow y = 9x + 15$$

equation must be seen in this form

A1

5

(b) (i) When $x = 2$, $y = 2^5 - 2 \times 2^2 + 9 = 32 - 8 + 9 = 33$

*be convinced that they are using **curve** equation*

$$(k =) 33$$

NMS $k = 33$ scores B0

B1

1



- (ii) When $x = 2$, $y = 9 \times 2 + 15 = 33$ so lies on tangent
*be convinced that they are using **tangent** equation
and have statement*

B1

1

(c) (i) $\frac{x^6}{6} - \frac{2x^3}{3} + 9x$

one of these terms correct

M1

another term correct

A1

all correct (may have +c)

A1

$$\left[\frac{2^6}{6} - \frac{2 \times 2^3}{3} + 9 \times 2 \right] - \left[\frac{(-1)^6}{6} - \frac{2 \times (-1)^3}{3} + 9 \times (-1) \right]$$

$$\left[\frac{64}{6} - \frac{16}{3} + 18 \right] - \left[\frac{1}{6} - \frac{2}{3} - 9 \right]$$

$$= 31.5$$

$F(2) - F(-1)$ unsimplified FT "their terms" from

integration = $\frac{70}{3} - \left(-\frac{49}{6} \right)$

m1

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$\left(\text{or } \frac{189}{6} \text{ etc} \right)$

condone single fraction

A1

5

- (ii) Area of trapezium = $\frac{1}{2} \times 3 \times (6 + \text{'their' } k)$
 $= 58.5$ when $k = 33$

B1 ✓

Shaded area = **Trapezium** – 'their' (c)(i) value

M1

$= 27$

OE eg $\frac{162}{6}$

A1

3

[15]

Q12.

(a) (i) $\{(2 + y)^3 =\} 8 + 12y + 6y^2 + y^3$

At least 3 terms simplified and correct

M1

All correct

A1

2

(ii) $(2 + x^{-2})^3 = 8 + 12x^{-2} + 6(x^{-2})^2 + (x^{-2})^3$

A replacement of y by x^{-2} in c 's (a)(i) working. PI

M1

$(2 - x^{-2})^3 = 8 - 12x^{-2} + 6(x^{-2})^2 - (x^{-2})^3$

Ft one incorrect coefficient in (a)(i) expansion.

A1F

$(2 + x^{-2})^3 + (2 - x^{-2})^3 = 16 + 12x^{-4}$

CSO Be convinced.

SC2 for a fully correct solution, not using 'Hence'

A1

3

(b) (i) $\int [(2 + x^{-2})^3 + (2 - x^{-2})^3] dx = 16x - 4x^{-3} (+c)$

Valid method to obtain the correct power of x after integrating qx^{-4} .

M1

$16x - 4x^{-3}$ or $16x - 4/x^3$ condone missing '+c'.

Ft on c 's p and q values. Coefficients and signs must be simplified

A1F

2

(ii) $\int_1^2 \dots dx = [16(2) - 4(2^{-3})] - [16 - 4]$

$F(2) - F(1)$ following integration (b)(i)

M1

$$= 31.5 - 12 = 19.5$$

OE Ft on c's **positive integer** values of p and q .

Since 'Hence' NMS scores 0/2

A1F

2

[9]

Q13.

(a) (i)

$$\left. \begin{array}{l} u = \ln x \quad \frac{dv}{dx} = 1 \\ \frac{du}{dx} = \frac{1}{x} \quad v = x \end{array} \right\}$$

$\frac{d \ln x}{dx}$ & $\int dx$ attempted

All correct

M1

A1

$$(\int \ln x \, dx =) \quad x \ln x - \int x \times \frac{1}{x} (dx)$$

Correct substitution of their terms into parts

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m1

$$= x \ln x - x + C$$

All correct (**constant needed**)

A1

4

(ii)

$$\left. \begin{array}{l} u = (\ln x)^2 \quad \frac{dv}{dx} = 1 \\ \frac{du}{dx} = (2 \ln x) \frac{1}{x} \quad v = x \end{array} \right\}$$

$$\frac{d(\ln x)^2}{dx} \text{ \& \int dx attempted}$$

M1

All correct

A1

$$\left(\int (\ln x)^2 dx = \right) x(\ln x)^2 - \int x \times \frac{2}{x} \ln x (dx)$$

OE correct substitution of their terms into parts

m1

$$= x(\ln x)^2 - 2(x \ln x - x) + C \quad \text{OE}$$

All correct (constant needed) including correct use of brackets. **Do not penalise missing constant if already penalised in part (i)**

ISW

A1

4

(b) $\frac{du}{dx} = \frac{1}{2\sqrt{x}} \quad \text{or} \quad \frac{1}{2}x^{-\frac{1}{2}}$

$$(dx = 2u \, du)$$

$$u = \sqrt{x}$$

B1

$$\int_{(1)}^{(4)} \frac{1}{x+\sqrt{x}} dx = \int_{(1)}^{(2)} \frac{1}{u^2+u} 2u \, (du)$$

All in terms of u including attempt at replacing dx (not simply writing du), condone missing limits and du

M1

Integrand correct unsimplified

A1

$$= 2 \int_{(1)}^{(2)} \frac{1}{u+1} (du)$$

A1

$$= 2 \ln(u+1) \Big|_{(1)}^{(2)}$$

FT their $\int \frac{k}{u+1} (du)$

A1F

$$= 2 \ln(2 + 1) - 2 \ln(1 + 1) \text{ or } 2 \ln(\sqrt{4} + 1) - 2 \ln(\sqrt{1} + 1)$$

correct use of correct limits on $k \ln(u + 1)$ or $k \ln(\sqrt{x} + 1)$

A1F

$$= 2 \ln \frac{3}{2} \text{ or } \ln \frac{9}{4} \text{ or } 2 \ln 3 - 2 \ln 2$$

OE ISW

A1

7

[15]

Q14.

(a) (i) $5 - 8x = A(1 - 3x) + B(2 + x)$

M1

$$x = -2 \quad x = \frac{1}{3}$$

Two values of x used to find values for A and B

m1

$$A = 3 \quad B = 1$$

A1

3

Alternative:

$$5 - 8x = A(1 - 3x) + B(2 + x)$$

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(M1)

$$5 = A + 2B$$

$$-8 = -3A + B$$

Set up simultaneous equations and solve.

(m1)

$$A = 3 \quad B = 1$$

(A1)

(3)

(ii) $\int_{-1}^0 \frac{3}{2+x} + \frac{1}{1-3x} \, dx = 3 \ln(2+x) - \frac{1}{3} \ln(1-3x)$



$a \ln(2+x) + b \ln(1-3x)$ where a and b are constants

M1

$$= (3 \ln 2 - \frac{1}{3} \ln 1) - (3 \ln 1 - \frac{1}{3} \ln 4)$$

$f(0) - f(-1)$ used

m1

$$= 3 \ln 2 + \frac{1}{3} \ln 4$$

ft A and B

A1ft

$$= \frac{11}{3} \ln 2$$

ft $\left(A + \frac{2}{3}B\right) \ln 2$

A1ft

4

(b) (i) $(C =)2$

B1

1

(ii) $\int \frac{9-18x-6x^2}{2-5x-3x^2} dx = \int C dx + \int \frac{5-8x}{2-5x-3x^2} dx$

Seen or implied.

Allow $\pm C + \int \frac{5-8x}{2-5x-3x^2} dx$

M1

$$\int_{-1}^0 \frac{9-18x-6x^2}{2-5x-3x^2} dx = 2 + \frac{11}{3} \ln 2$$

Accept $2 + 3 \ln 2 + \frac{1}{3} \ln 4$

ft 2 + candidate's answer to part (a)(ii) if exact.

A1ft

2

[10]

Q15.

(a) (i) $\left(\frac{dy}{dx} = \right) 5x^4 - 6x + 1$

one term correct

M1



another term correct

A1

all correct (no + c etc)

A1

(ii) $\left(\frac{d^2y}{dx^2}\right) 20x^3 - 6$
 FT 'their' $\frac{dy}{dx}$

B1✓

(b) $x = -1 \Rightarrow \frac{dy}{dx} = 5(-)^4 - 6(-1) + 1 (= 12)$
 must sub $x = -1$ into 'their' $\frac{dy}{dx}$

M1

$\Rightarrow y = 12(x + 1)$

any correct form with $(x - -1)$ simplified
 condone $y = 12x + c$, $c=12$

A1cso

(c) $x = 1 \Rightarrow \frac{dy}{dx} = 5 - 6 + 1$
 sub $x = 1$ into their $\frac{dy}{dx}$

$\frac{dy}{dx} = 0 \Rightarrow$ stationary point

shown = 0 plus correct statement

A1cso

when $x = 1$, $\frac{d^2y}{dx^2} = 14 \Rightarrow (B \text{ is a })$ minimum (point)

or $\frac{d^2y}{dx^2} = 20 - 6 > 0 \Rightarrow (B \text{ is a })$ minimum (point)

must have correct $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for E1

E1

(d) (i) $\frac{x^6}{6} - \frac{3x^3}{3} + \frac{x^2}{2} + 5x$
 one term correct

M1

another term correct

A1

all correct (may have + c)

A1

$$\left[\frac{1}{6} - 1 + \frac{1}{2} + 5 \right] - \left[\frac{1}{6} + 1 + \frac{1}{2} - 5 \right]$$

'their' $F(1) - F(-1)$ with powers of 1 and -1 evaluated correctly

M1

$$= 8$$

A1cso

5

(ii) 'their answer to part (i)' - 2

M1

$$\Rightarrow \text{Area} = 6$$

A1cso

2

[16]

Q16.

(a) $\frac{dy}{dx} = 12 - 5x^{\frac{2}{3}}$
 $kx^{\frac{2}{3}}$ term.

M1

ACF

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = 12$

Ft on c's y' evaluated correctly at $x = 0$

B1F

Eqn of tangent at O is $y = 12x$

OE Ft on c's value for $y'(0)$ provided $y'(0) > 0$.

B1F

2

(ii) When $x = 8$, $\frac{dy}{dx} = 12 - 5 \times (8)^{\frac{2}{3}}$

Attempt to find $\frac{dy}{dx}$ when $x = 8$

M1

Equation of tangent at (8, 0) is $y - 0 = y'(8)[x - 8]$

$y = y'(8)[x - 8]$ OE

M1

$y = -8(x - 8) \Rightarrow y + 8x = 64$

CSO AG

A1

3

(c) $\int \left(12x - 3x^{\frac{5}{3}} \right) dx = \frac{12x^2}{2} - \frac{3x^{\frac{8}{3}}}{\frac{8}{3}} (+c)$

$kx^{\frac{5}{3}+1}$ term after integrating, condone k left unsimplified for this M mark.

$= 6x^2 - \frac{9}{8}x^{\frac{8}{3}} (+c)$

For $6x^2$ OE eg $(12x^2 / 2)$

B1

For $-\frac{9}{8}x^{\frac{8}{3}}$ OE

A1

3

(d) Area bounded by curve and x -axis

$= \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx = 6 \times 8^2 - \frac{9}{8} \times (8)^{\frac{8}{3}}$

$\pm F(8) \{- F(0)\}$ PI following integration

M1

$= 384 - 288 = 96$

PI by correct final answer if evaluation not seen here

A1

At P , $12x + 8x = 64$

Solving $y + 8x = 64$ and c's $y = kx$, $k > 0$, down to an eqn in one variable...

$[y + 2y / 3 = 64]$

M1

$(x_p = 3.2) \quad y_p = 38.4$

For $y_p 38.4$ OE [If using integration to find area of triangle, award A1 if both ' $x_p = 3.2$ ' and correct integration of correct eqns of the 2 lines]

A1

Area of triangle $OPA = \frac{1}{2} \times 8 \times y_p$

OE Need perpendicular ht to be linked to $y_p > 0$.

M1

Area of shaded region = Area $\Delta OPA - \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx$
M0 if evaluated to a value < 0

M1

$= 153.6 - 96 = 57.6$

OE eg 288 / 5

A1

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7

[17]

Q17.

(a) $\left(\frac{dy}{dx} = \right) 12x^2 - 6$
do not ISW

B1

1

(b) $\int_2^3 \frac{2x^2 - 1}{4x^3 - 6x + 1} dx = \left[\frac{1}{6} \ln(4x^3 - 6x + 1) \right]_{(2)}^{(3)}$
 $k \ln(4x^3 - 6x + 1)$, k is a constant

M1

$$k = \frac{1}{6}$$

A1

$$= \frac{1}{6} \ln(4 \times 3^3 - 6 \times 3 + 1)$$

$$- \frac{1}{6} \ln(4 \times 2^3 - 6 \times 2 + 1)$$

correct substitution in $F(3) - F(2)$.
condone poor use or lack of brackets.

m1

$$= \frac{1}{6} \ln 91 - \frac{1}{6} \ln 21$$

$k \ln 91 - k \ln 21$
only follow through on their k

A1F

$$= \frac{1}{6} \ln \frac{91}{21} \quad \text{or} \quad \left(= \frac{1}{6} \ln \frac{13}{3} \right)$$

A1

or if using the substitution

$$u = 4x^3 - 6x + 1$$

$$\int = k \int \frac{du}{u} \quad \text{M1}$$

$$= \frac{1}{6} \ln u \quad \text{A1}$$

then, either change limits to 21 and 91 m1 then A1F A1
as scheme or changing back to 'x', then m1 A1F A1 as
scheme

5

[6]

Q18.

$$(a) \quad \left(\frac{dx}{d\theta} = \right) \frac{(\sin \theta \times 0) - 1 \times \cos \theta}{\sin^2 \theta}$$

$$\text{quotient rule} \quad \frac{\pm \sin \theta \times k \pm 1 \times \cos \theta}{\sin^2 \theta} \quad \text{where } k = 0 \text{ or } 1$$

M1

must see the '0' either in the quotient or in eg $\frac{du}{d\theta} = 0$ etc

A1



$$= -\frac{\cos \theta}{\sin^2 \theta} \quad \text{or} \quad = -\frac{\cos \theta}{\sin \theta \sin \theta}$$

or equivalent

$$= -\operatorname{cosec} \theta \cot \theta$$

CSO, AG must see one of the previous expressions

A1

(b) $x = \operatorname{cosec} \theta$

$$\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$$

OE, eg $dx = -\operatorname{cosec} \theta \cot \theta d\theta$

B1

Replacing $\sqrt{(\operatorname{cosec}^2 \theta - 1)}$ by $\sqrt{\cot^2 \theta}$, or better
at any stage of solution

B1

$$\int = \int \frac{-\operatorname{cosec} \theta \cot \theta}{\operatorname{cosec}^2 \theta \sqrt{(\operatorname{cosec}^2 \theta - 1)}} d\theta$$

all in terms of θ , and including their attempt at dx ,
but condone omission of $d\theta$

M1

fully correct and must include $d\theta$ (at some stage in solution)

A1

$$\int \frac{-\operatorname{cosec} \theta \cot \theta}{\operatorname{cosec}^2 \theta \cot \theta} (d\theta) = \int \frac{-1}{\operatorname{cosec} \theta} (d\theta)$$

OE eg $\int -\sin \theta (d\theta)$

A1

$$= \cos \theta$$

A1

$$x = 2, \theta = 0.524 \text{ AWRT}$$

$$x = \sqrt{2}, \theta = 0.785 \text{ AWRT}$$

$$\text{correct change of limits or } (\pm) \cos \theta = (\pm) \left[\sqrt{1 - \frac{1}{x^2}} \right]_{\sqrt{2}}^2 \quad \text{OE}$$

B1



$$0.8660 - 0.7071$$

$$c's F(0.52) - F(0.79)$$

$$\text{substitution into } \pm \cos\theta \text{ only or } \left(\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \right)$$

m1

$$= 0.159$$

A1

9

[12]

Q19.

$$(a) \quad 2x + 3 = A(2x + 1) + B(2x - 1)$$

M1

$$x = \frac{1}{2} \quad x = -\frac{1}{2}$$

Use two values of x to find A and B

m1

$$A = 2 \quad B = -1$$

Both

A1

3

Alternative; equating coefficients

$$2x + 3 = A(2x + 1) + B(2x - 1)$$

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(M1)

$$x \text{ term} \quad 2 = 2A + 2B$$

$$\text{constant} \quad 3 = A - B$$

Set up simultaneous equations and solve.

(m1)

$$A = 2 \quad B = -1$$

Both

(A1)

(3)

Alternative; cover up rule

$$x = \frac{1}{2} \quad A = \frac{2 \times \frac{1}{2} + 3}{2 \times \frac{1}{2} + 1} \quad \left(= \frac{4}{2} \right)$$

$$x = \frac{1}{2} \quad \text{and} \quad x = -\frac{1}{2} \quad \text{used to find } A \text{ and } B$$

(M1)

$$x = -\frac{1}{2} \quad B = \frac{2 \times (-\frac{1}{2}) + 3}{2 \times (-\frac{1}{2}) - 1} \quad \left(= \frac{2}{-2} \right)$$

SC NMS

$$A = 2 \quad B = -1$$

A and B both correct 3 / 3

One of A or B correct 1 / 3

(A1A1)

(3)

Condone poor algebra for M1 if continues correctly.

(b)

$$\begin{array}{r}
 3x \\
 4x^2 - 1 \overline{) 12x^3 - 7x - 6} \\
 \underline{12x^3 - 3x} \\
 -4x - 6
 \end{array}$$

Complete division leading to values for C and D

M1

$$C = 3$$

EXAM PAPERS PRACTICE

A1

$$D = -2$$

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A1

C = 3, D = -2 stated or written in expression.

SC B1

C = 3, D not found or wrong;

D = -2, C not found or wrong.

3

Alternative

$$\frac{12x^3 - 7x - 6}{4x^2 - 1} = \frac{12x^3 - 3x - 4x - 6}{4x^2 - 1} = 3x - \frac{2(2x+3)}{4x^2 - 1}$$

(M1)

$$C = 3$$

(A1)

$$D = -2$$

(A1)

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2, C$ not found or wrong.

(3)

Alternative

$$12x^3 - 7x - 6 = 4Cx^3 - Cx + 2Dx + 3D$$

Complete method for C and D

(M1)

$$C = 3$$

(A1)

$$D = -2$$

(A1)

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2, C$ not found or wrong.

(3)

Alternative

$$x = 0$$

$$x = 1$$

$$6 = -3D \quad -\frac{1}{3} = C + \frac{5}{3}D$$

Use two values of x to set up simultaneous equations

(M1)

$$C = 3$$

(A1)

$$D = -2$$

(A1)

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2$, C not found or wrong.

(3)

Complete division for M1; obtain a value for $C(Cx)$ and a remainder $ax + b$

(c) $\int 3x - 2 \left(\frac{2}{2x-1} - \frac{1}{2x+1} \right) dx$

Use parts (a) and (b) to obtain integrable form

M1

$$3 \frac{x^2}{2}$$

ft on C

A1ft

$$-2 \left(\ln(2x-1) - \frac{1}{2} \ln(2x+1) \right)$$

Both correct; ft on A , B and D

Condone missing brackets

A1ft

$$\frac{3}{2}(4-1) - 2 \left(\left(\ln 3 - \frac{1}{2} \ln 5 \right) - \left(\ln 1 - \frac{1}{2} \ln 3 \right) \right)$$

Correct substitution of limits

m1

$$\frac{9}{2} - 3 \ln 3 + \ln 5 = \frac{9}{2} + \ln \left(\frac{5}{27} \right)$$

$$p = \frac{9}{2} \quad q = \frac{5}{27}$$

A1

Form $\int Cx + \left(\frac{P}{2x-1} + \frac{Q}{2x+1} \right) dx$ using candidate's P , Q , C for M1.

Condone missing dx .

$$\int Cx \, dx = C \frac{x^2}{2} \text{ for A1ft}$$

ISW extra terms eg $\frac{12}{4x^2-1}$ for first three terms only; max 3/5.

Candidate's C ; must have a value.

$$\int \frac{4x+6}{4x^2-1} dx = \int \frac{4x}{4x^2-1} + \frac{6}{4x^2-1} dx \text{ is an integrable form,}$$

as $\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \left(\frac{x-a}{x+a} \right)$ is in the formula book, but they **must** try to

integrate to show they know this, **or** use partial fractions again with

$$\frac{6}{4x^2-1} = \frac{3}{2x-1} - \frac{3}{2x+1} \text{ for M1.}$$

Substitute limits into $C \frac{x^2}{2} + m \ln(2x-1) + n \ln(2x+1)$, or equivalent, for m1;
substitution must be completely correct.

Condone $\frac{9}{2} - \ln \left(\frac{27}{5} \right)$ for A1.

5

[11]

Q20.

(a) (i) $\left(x - \frac{3}{2}\right)^2$
or $p = 1.5$ stated

EXAM PAPERS PRACTICE M1

$$\left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$$

$$(x - 1.5)^2 + 2.75$$

A1

Mark their final line as their answer

2

(ii) $x = \frac{3}{2}$

correct or FT their " $x = p$ "

B1 ✓

1

(b) (i) $x^2 - 3x + 5 = x + 5 \Rightarrow x^2 = 4x$

eliminating x or y and collecting like terms

(condone **one** slip)

or $(y - 5)^2 - 3(y - 5) + 5 = y \Rightarrow y^2 - 14y + 45 = 0$

M1

$(x \neq 0) \Rightarrow x = 4$

A1

$y = 9$

A1

3

(ii) $\frac{x^3}{3} - \frac{3x^2}{2} + 5x (+c)$

one of these terms correct

M1

another term correct

A1

all correct (need not have + c)

A1

3

(iii) $\left[\right]_0^4 = \frac{4^3}{3} - 3 \times \frac{4^2}{2} + 5 \times 4$

must have earned M1 in part(b)(ii)

$F(\text{their } x_B) \{-F(0)\}$ "correctly sub'd"

M1

$= 17\frac{1}{3}$

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$\left(\frac{64}{3} - 24 + 20 = \right) \frac{52}{3} \text{ or } \frac{104}{6} \text{ etc}$

condone 17.3 but not $16\frac{4}{3}$ etc

A1

Area trapezium $\frac{1}{2}(x_B)(5 + y_B)$

FT their numerical values of x_B, y_B

Area = $\frac{1}{2} \times 4 \times 14 (= 28)$

B1 ✓

Area of shaded region $28 - 17\frac{1}{3}$

$$= 10\frac{2}{3}$$

CSO; $\frac{32}{3}$, accept 10.7 or better

A1

4

[13]

Q21.

(a) = $\left(x^{\frac{3}{2}}\right)^2 - 2x^{\frac{3}{2}} + 1 = x^3 - 2x^{\frac{3}{2}} + 1$

B2 for $x^3 - 2x^{\frac{3}{2}} + 1$ or $x^3 - 2x\sqrt{x} + 1$
(B1 fully correct unsimplified expression.

seen eg $\left(x^{\frac{3}{2}}\right)^2 - x^2 - x^{\frac{3}{2}} + 1$

or B1 for either $x^3 + 2x^{\frac{3}{2}}$... OE seen

or $x^3 + 2x^{\frac{3}{2}} - 1$ OE seen)

or B1 for $-x^3 + 2x^{\frac{3}{2}} - 1$ OE seen)

B2,1,0

2

(b) $\int \left(x^{\frac{3}{2}} - 1\right)^2 dx = \frac{x^4}{4} - \frac{2x^{\frac{5}{2}}}{2.5} + x (+c)$

Ft on correct integration of all non $x^{\frac{3}{2}}$ terms
(at least two) in c's expression. in (a)

B1F

Integration of a $kx^{\frac{3}{2}}$ as $\lambda x^{\frac{5}{2}}$ (ie power correct)

M1

$$\{= 0.25x^4 - 0.8x^{2.5} + x (+c)\}$$

Correct integration of c's $x^{\frac{3}{2}}$ term(s)
ACF

A1F

3

$$(c) \int_1^4 \left(x^{\frac{3}{2}} - 1 \right)^2 dx = \left(\frac{4^4}{4} - \frac{2(4^{\frac{3}{2}})}{2.5} + 4 \right) - \left(\frac{1}{4} - \frac{2}{2.5} + 1 \right)$$

F(4) – F(1) attempted following integration. If F(x) incorrect, ft c's answer to (b) provided integration attempted

M1

$$\{ = \frac{212}{5} - \frac{9}{20} = 42.4 - 0.45 \} = 41.95$$

41.95 OE eg 839 / 20

Since 'Hence' NMS scores 0 / 2

A1

2

[7]

Q22.

$$u = x^4 + 2$$

$$\frac{du}{dx} = 4x^3$$

$$\text{or } du = 4x^3 dx$$

B1

$$\int \frac{x^7}{(x^4 + 2)^2} dx = \int \frac{k(u-2)}{u^2} du \text{ or } \int \frac{k(u-2)^{\frac{7}{4}}}{u^2} \frac{du}{(u-2)^{\frac{3}{4}}}$$

Either expression all in terms of u including replacing dx, but condone omission of du

M1

$$= \left(\frac{1}{4} \right) \int \frac{1}{u} - \frac{2}{u^2} du$$

$$k \int au^{-1} + bu^{-2} du, \text{ where } k, a, b \text{ are constants}$$

m1

$$= \left(\frac{1}{4} \right) \left[\ln u + \frac{2}{u} \right]$$

Must have seen du on an earlier line where every term is a term in u

A1

$$= \left(\int = \left(\frac{1}{4} \right) \left[\ln u + \frac{2}{u} \right]_2^3 \right)$$

$$\left(\left(\frac{1}{4} \right) \left[\ln(x^4 + 2) + \frac{2}{(x^4 + 2)} \right]_0^1 \right)$$

$$= \left(\frac{1}{4} \right) \left[\left(\ln 3 + \frac{2}{3} \right) - (\ln 2 + 1) \right]$$

Dependent on previous A1

Correct change of limits, correct substitution and $F(3) - F(2)$

or

correct replacement of u , correct substitution and $F(1) - F(0)$

m1

$$= \frac{1}{4} \ln \left(\frac{3}{2} \right) - \frac{1}{12}$$

OE in exact form

A1

[6]

Q23.

(a) (i) $\frac{dy}{dx} = -1 - 4x^3$

one of these terms correct

M1

all correct (no + c)

A1

(When $x = 1$, grad =) -5

(Check that $\frac{dy}{dx}$ is actually correct!)

A1cso

3

(ii) $y - 12 = \text{'their grad'} (x - 1)$

any form of equation through (1, 12) and

attempt at c if using $y = mx + c$

M1

$y = -5x + 17$ (or $y = 17 - 5x$)

FT their gradient

Condone $y = -5x + c$, $c = 17$ etc

A1 ✓

2

(b) (i) $14x - \frac{x^2}{2} - \frac{x^5}{5}$

one of these terms correct

M1

another term correct

A1

all correct (may have + c)

A1

$[]_2 =$

$\left(14 - \frac{1}{2} - \frac{1}{5}\right) - \left(-28 - 2 + \frac{32}{5}\right)$

F(1) and F(-2) attempted

m1

$= 36.9$ OE

Condone recovery to this value

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 3 \times 12$

Correct area of triangle unsimplified

M1

$= 18$

$\Rightarrow$ shaded area = 18.9

A1cso

2

[12]

Q24.

(a) $y = x + 3 + \frac{8}{x^4} = x + 3 + 8x^{-4}$

For $\frac{8}{x^4} = 8x^{-4}$ PI by correct differentiation

of 3rd term

B1

$$\frac{dy}{dx} = 1 - 32x^{-5} \text{ or } 1 - \frac{32}{x^5}$$

$$k x^{-5} \quad \text{OE}$$

M1

For either

A1

3

(b) When $x = 1, y = 12$

B1

When $x = 1, \frac{dy}{dx} = 1 - 32 = -31$

Attempt to find value of $\frac{dy}{dx}$ when $x = 1$

M1

Tangent: $y - 12 = -31(x - 1)$

Only ft on c's answer to (a). Any correct (ft on c's (a)) form.

A1F

3

(c) $1 - 32x^{-5} = 0$

$1 - 32x^{-5} = 0$ or c's $\frac{dy}{dx} = 0$

M1

$$\Rightarrow x^5 = 32$$

Attempt to form $x^n = \text{const} (\neq 0)$. Pl by next line

m1

$$\Rightarrow x = 2$$

CSO

A1

(Coordinates of M) (2, 5.5)

CSO

A1

4

(d) (i)
$$\int \left(x + 3 + \frac{8}{x^4} \right) dx$$

$$= \frac{x^2}{2} + 3x - \frac{8}{3}x^{-3} + c$$

Power -3 correctly obtained

M1

$$- \frac{8}{3}x^{-3}$$

A1

$$\frac{x^2}{2} + 3x + c$$

B1

3

(ii)

$$= \left(2 + 6 - \frac{1}{3} \right) - \left(\frac{1}{2} + 3 - \frac{8}{3} \right)$$

Attempting to calculate $F(2) - F(1)$ where $F(x)$ is c's answer to part (d)(i) provided F is not just the c's integrand $(x + 3 + 8/x^4)$

M1

$$\frac{9}{2} + \frac{7}{3} = \frac{41}{6}$$

OE Accept 6.83 or better provided d(i) used

A1

2

(e) $k = -5.5$

Ft on $-y_M$ from part (c).

B1F

1

[16]

Q25.

(a) $\int_{-1}^1 (x^3 - 2x^2 + 3) dx$

$$= \left[\frac{x^4}{4} - \frac{2x^3}{3} + 3x \right]_{-1}^1$$

one term correct

M1

another term correct

A1

all correct (condone + c)

A1

$$= \left(\frac{1}{4} - \frac{2}{3} + 3 \right) - \left(\frac{1}{4} + \frac{2}{3} - 3 \right)$$

'their' $F(1) - F(-1)$

with $(-1)^3$ etc evaluated correctly

but must have earned M1

B1 ✓

$$= 4 \frac{2}{3}$$

$$\frac{14}{3}, \frac{56}{12} \text{ etc.}$$

but combined as single fraction

A1cso

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5

(b) Area of $\Delta \left(= \frac{1}{2} \times 2 \times 2 \right)$

$$= 2$$

PI

B1

Shaded region has area $4 \frac{2}{3} - 2$

±their (a) ±their Δ area

M1

$$= 2\frac{2}{3}$$

$$\frac{8}{3}, \frac{32}{12} \text{ etc.}$$

but combined as single fraction

A1cso

3

[8]

Q26.

(a) $(2 + x^2)^3$

$$= [(2)^3] + 3(2)^2(x^2) + 3(2)(x^2)^2 + (x^2)^3$$

For either (1), 3, 3, (1) OE unsimplified

$$\text{or } \binom{3}{1} 2^2 x^2 + \binom{3}{2} 2(x^2)^2 \text{ OE. PI}$$

M1

$$p = 3(2)^2 = 12$$

AG Be convinced. Condone left as $12x^2$

A1

$$q = 6$$

Accept left as $6x^4$

B1

3

(b) (i) $\int \frac{(2 + x^2)^3}{x^4} dx = \int x^{-4} (8 + 12x^2 + 6x^4 + x^6) dx$

$$\text{or } \int \left(\frac{8}{x^4} + \frac{12}{x^2} + 6 + x^2 \right) dx$$

$$\frac{1}{x^n} = x^{-n}$$

Uses (a) and either an indication that
in a product PI or cancelling to get at
least 3 correct ft terms

M1

$$\int (8x^{-4} + 12x^{-2} + 6 + x^2) dx$$

Ft on c's non-zero q. PI by next line in solution

A1F

$$= \frac{8x^{-3}}{-3} + \frac{12x^{-1}}{-1} + qx + \frac{x^3}{3} \{+c\}$$

*Correct integration of either $8x^{-4}$ or $12x^{-2}$;
accept unsimplified*

M1

*Correct integration of both $8x^{-4}$ and $12x^{-2}$;
accept unsimplified coefficients*

A1

$$= \dots\dots\dots + 6x + \frac{x^3}{3} + c$$

$$(= -\frac{8}{3}x^{-3} - 12x^{-1} + 6x + \frac{x^3}{3} + c)$$

*For "6" $x + \frac{x^3}{3} + c$ simplified. The only ft is "6"
replaced by c's value for q where q is a
non-zero integer.*

B1F

5

$$(ii) \int_1^2 \frac{(2+x^2)^3}{x^4} dx = \left\{ -\frac{8}{3}(2)^{-3} - 12(2)^{-1} + 6(2) + \frac{2^3}{3} \right\} -$$

$$\left\{ -\frac{8}{3}(1)^{-3} - 12(1)^{-1} + 6(1) + \frac{1^3}{3} \right\}$$

*Dealing correctly with limits; $F(2)-F(1)$ (must
have attempted integration to get F ie c's F is
not just the integrand)*

M1

$$= \left(-\frac{1}{3} - 6 + 12 + \frac{8}{3} \right) - \left(-\frac{8}{3} - 12 + 6 + \frac{1}{3} \right)$$

$$= 16\frac{2}{3}$$

*OE exact answer eg 50/3.
NMS scores 0*

A1

2

Q27.

(a) (i) $\left(\frac{dx}{d\theta}\right) = -6\sin 2\theta, \quad \left(\frac{dy}{d\theta}\right) = -2\sin\theta$

$$\left(\frac{dx}{d\theta}\right) = p\sin\theta \text{ or } r\sin\theta\cos\theta$$

$$\left(\frac{dy}{d\theta}\right) = q\sin\theta$$

M1

Both correct.

A1

$$\frac{dy}{dx} = \frac{-2\sin\theta}{-6\sin 2\theta}$$

$$\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Use chain rule ; condone one slip

M1

$$= \frac{2\sin\theta}{6 \times 2\sin\theta\cos\theta} = \frac{1}{6\cos\theta}$$

k = 6 must come from correct working seen **AG**

A1

4

(ii) $\theta = \frac{\pi}{3} \quad m_T = \frac{1}{3}$

$$\left(\frac{1}{k \times \frac{1}{2}} \right)$$

ft on k

k need not be numerical

B1ft

$$m_N = -3$$

ft on m_T

B1ft

$$(x, y) = \left(-\frac{3}{2}, 1\right)$$

B1

Normal $y - 1 = -3\left(x + \frac{3}{2}\right)$

CAO; any correct form, ISW.

$$2y + 6x + 7 = 0$$

B1

4

(b) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$p + q\cos 2x$; Allow different letters for x or mixture eg θ even for A1 and the following A1ft

M1

A1

$$\int p \, dx = px \quad \int q \cos 2x = \frac{1}{2} q \sin 2x$$

Both integrals correct;
ft on p and q

A1ft

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x \, dx = \left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$$

$$= \left(\frac{\pi}{8} - \frac{1}{4} \right) - \left(-\frac{\pi}{8} - \left(-\frac{1}{4} \right) \right)$$

Correct use of limits;

$$F\left(\frac{\pi}{4}\right) - F\left(-\frac{\pi}{4}\right) \text{ or } 2F\left(\frac{\pi}{4}\right)$$

$$F(x) = px + r\sin 2x \text{ and } \sin \frac{\pi}{2}, \sin\left(-\frac{\pi}{2}\right)$$

must be evaluated correctly for m1

m1

$$= \frac{\pi}{4} - \frac{1}{2}$$

CSO OE ISW

A1

5

Alternative

$$\int \sin^2 x dx = -\sin x \cos x - \int -\cos x \cos x dx$$

Use parts; condone sign slips

(M1)

$$= -\sin x \cos x + \int 1 - \sin^2 x dx$$

Use $\cos^2 x = 1 - \sin^2 x$

(m1)

$$2 \int \sin^2 x dx = -\sin x \cos x + x$$

(A1)

$$2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = G\left(\frac{\pi}{4}\right) - G\left(-\frac{\pi}{4}\right)$$

Correct use of limits

(m1)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \frac{\pi}{4} - \frac{1}{2}$$

(A1)

(5)

[13]

Q28.

(a) (i) $\frac{dy}{dx} = 24x - 19 - 6x^2$

2 terms correct

M1

all correct (no + c etc)

A1

when $x = 2, \frac{dy}{dx} = 48 - 19 - 24$

m1

$\Rightarrow$ gradient = 5
CSO

A1

4

(ii) grad of normal = $-\frac{1}{5}$
ft their answer from (a)(i)

B1ft

$y + 6 = \left(\text{their} - \frac{1}{5} \right) x - 2$
 or $y = \left(\text{their} - \frac{1}{5} \right) x + c$ and c evaluated
*ft grad of their normal using **correct** coordinates BUT must not be tangent*
condone omission of brackets

M1

using $x = 2$ and $y = -6$
 $x + 5y + 28 = 0$
 CSO; condone all on one side in different order

A1

3

(b) (i) $\frac{12}{3}x^3 - \frac{19}{2}x^2 - \frac{2}{4}x^4$
one term correct

M1

another term correct

A1

all correct (ignore + c or limits)

A1

$$= 32 - 38 - 8$$

F 2 attempted

m1

$$= -14$$

CSO; withhold A1 if changed to +14 here

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 2 \times 6 = 6$

condone -6

B1

Shaded region area = $14 - 6$

difference of $\pm$ | $\int$ | $\pm$ | Δ |

$$= 8$$

CSO

A1

3

[15]

Q29.

(a) (i) $\frac{x^5}{5} - \frac{8}{2}x^2 + 9x$

One term correct

M1

Another term correct

A1

All correct (may have + c)

A1

$$\frac{32}{5} - 16 + 18$$

F(2) attempted

m1

$$= 8\frac{2}{5}$$

$$\frac{42}{5}, 8.4$$

A1

5

- (ii) Shaded area = $18 - \text{'their integral'}$
PI by 18 – (a)(i) NMS

M1

$$= 9\frac{3}{5}$$

$$\frac{48}{5}, 9.6 \text{ NMS full marks}$$

A1

2

(b) (i) $\frac{dy}{dx} = 4x^3 - 8$

One term correct

M1

All correct (no + c etc)

A1

$$x = 1 \Rightarrow \frac{dy}{dx} = 4 - 8$$

sub $x = 1$ into their $\frac{dy}{dx}$

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m1

(Gradient of curve) = -4
No ISW

A1 cso

4

- (ii) $y - 2 = -4(x - 1); y = -4x + c, c = 6$
any correct form; FT their answer from (b)(i) but must use $x = 1$ and $y = 2$

B1ft

1

[12]

Q30.

(a) $p = -3; q = 3$

Accept even if just embedded in the expansion

B1;B1

2

(b) (i) $\int \left(1 - \frac{1}{x^2}\right)^3 dx = \int (1 - 3x^{-2} + 3x^{-4} - x^{-6}) dx$

Uses (a) with indication of integration and

indication of $\frac{1}{x^n} = x^{-n}$ PI

M1

$$= x + 3x^{-1} - x^{-3} + \frac{1}{5}x^{-5} \{+c\}$$

At least three powers of x correctly obtained

m1

*Ft on c's non-zero integers p and q.
A1F if 3 of the 4 terms are correct (ft) or
if all correct (ft) but left unsimplified
Condone missing +c.*

A2F,1F

4

(ii) $\int_{\frac{1}{2}}^1 \left(1 - \frac{1}{x^2}\right)^3 dx = \left(1 + 3 - 1 + \frac{1}{5}\right) - \left(\frac{1}{2} + 6 - 8 + \frac{32}{5}\right)$

*Attempting to calculate $F(1) - F(1/2)$
where F is c's answer to part (b)(i)
provided F is not the integrand or the c's*

equivalent of the integrand $\left(1 - \frac{1}{x^2}\right)^3$

M1

$$= -\frac{17}{10}$$

OE exact answer eg -1.7

A1

2

[8]

Q31.

(a) $y = e^x \rightarrow e^{2x} - 1$

Stretch (I)

scale factor $\frac{1}{2}$ (II)

I + (II or III)

M1

in x -direction (III)

I + II + III

A1

Translation

Allow "translate"

E1

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

OE "1 unit down" etc

B1

4

(b) $x = 0$ $y = 6$ or $(0, 6)$

Both coordinates must be stated, not simply 6 marked on diagram

B1

1

(c) (i) $e^{2x} - 1 = 4e^{-2x} + 2$

$$e^{4x} - e^{2x} = 4 + 2e^{2x}$$

$$\text{or } (e^{2x})^2 - e^{2x} = 4 + 2e^{2x}$$

Multiplying both sides by e^{2x}

M1

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

AG With no errors seen

A1

2

(ii) $(e^{2x} - 4)(e^{2x} + 1)$

$$(e^{2x} \pm 4)(e^{2x} \pm 1)$$

M1

$$x = \ln 2 \text{ or } \frac{1}{2} \ln 4$$

A1

Reject $e^{2x} = -1$ OE

Eg $e^{2x} > 0$, $e^{2x} \neq -1$, impossible etc

A1

3

(d) $\int (4e^{-2x} + 2) dx$ (I)

$$= \left[\frac{4e^{-2x}}{-2} + 2x \right]_0^{\ln 2}$$

I or II attempted and e^{-2x} or e^{2x} integrated correctly

M1

$$= \left(\frac{4e^{-2\ln 2}}{-2} + 2\ln 2 \right) - \left(\frac{4}{-2} + 0 \right)$$

F [‘their $\ln 2$ ’ from (c)(ii)] – F[0]

m1

$$= -\frac{1}{2} + 2\ln 2 + 2 = \frac{3}{2} + 2\ln 2$$

$$\int (e^{2x} - 1) dx$$
 (II)

$$= \left[\frac{e^{2x}}{2} - x \right]_0^{\ln 2}$$

Both I and II correctly integrated

A1

$$= \left(\frac{e^{2\ln 2}}{2} - \ln 2 \right) - \left(\frac{1}{2} - 0 \right)$$

$$= 2 - \ln 2 - \frac{1}{2} = \frac{3}{2} - \ln 2$$

$$A = \left(\frac{3}{2} + 2\ln 2 \right) - \left(\frac{3}{2} - \ln 2 \right)$$

Attempt to find difference of ‘their I – their II’

B1ft



$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

A1

Alternative

$$A = \int (4e^{-2x} + 2) dx - \int (e^{2x} - 1) dx$$

Condone functions reversed

(B1)

$$= \int_{(0)}^{(\ln 2)} (4e^{-2x} - e^{2x} + 3) dx$$

$$= \left[\frac{4e^{-2x}}{-2} - \frac{e^{2x}}{2} + 3x \right]_0^{\ln 2}$$

e^{2x} or e^{-2x} correctly integrated

(M1)(A1)

$$= \left(-2e^{-2 \ln 2} - \frac{1}{2}e^{2 \ln 2} + 3 \ln 2 \right) - \left(-2 - \frac{1}{2} \right)$$

Correct substitution of their ln 2 from (c)(ii) into their integrated expression

(m1)

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

(A1)

5

[15]

Q32.

(a) (i) $p(2) = 8 + 2 - 10$

Must find p(2) NOT long division

M1

$$\Rightarrow p(2) = 0 \Rightarrow (x - 2) \text{ is factor}$$

Shown = 0 plus a statement

A1

2

- (ii) Attempt at long division (generous)

*Obtaining a quotient $x^2 + cx + d$
or equating coefficients (full method)*

M1

$$p(x) = (x - 2)(x^2 + 2x + 5)$$

a = 2, b = 5 by inspection B1, B1

A1

2

(b) (i) $\frac{dy}{dx} = 3x^2 + 1$

One term correct

M1

All correct – no + c etc

A1

When $x = 2 \quad \frac{dy}{dx} = 3 \times 4 + 1$

Sub $x = 2$ into their $\frac{dy}{dx}$

m1

Therefore gradient at Q is 13

CSO

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A1

4

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- (ii) $y = 13(x - 2)$

*Tangent (NOT normal) attempted
ft their gradient answer from (b)(i)*

M1

CSO; correct in any form

A1

2

(iii) $\int \dots dx = \frac{x^4}{4} + \frac{x^2}{2} - 10x (+c)$

one term correct

M1

second term correct

A1

all correct (condone no +c)

A1

3

(iv) $[4 + 2 - 20] - [0] = -14$

F(2) attempted and possibly F(0)

Must have earned M1 in (b)(iii)

M1

Area of shaded region = 14

CSO; separate statement following correct evaluation of limits

A1

2

[15]

Q33.

(a) (i) $\frac{dy}{dx} = 3x^{\frac{1}{2}}$
 $kx^{\frac{1}{2}}$ with or without + c

M1

= 6 {when $x = 4$ }

Must be 6 and seen in (a)(i)

6 + c is A0

A1cao

2

(ii) y -coordinate of A = $2 \times 4^{\frac{3}{2}} (= 16)$

Substitute $x = 4$ in $y = 2x^{\frac{3}{2}}$

M1

$6 \times m' = -1$

$m_1 \times m_2 = -1$ OE used with c's value of

$\frac{dy}{dx}$ when $x = 4$. PI

M1



$$y - 16 = m(x - 4)$$

dep on 1st M1 in (a)(ii)

m must be numerical

m1

$$y - 16 = -\frac{1}{6}(x - 4)$$

ACF

A1

4

(b) (i)

$$\int 8x^{\frac{1}{2}} dx = \frac{8}{\frac{3}{2}} x^{\frac{1}{2}+1} \{+c\}$$

Index raised by 1

M1

$$= \frac{16}{3} x^{\frac{3}{2}} \{+c\}$$

Condone missing '+ c'
Coefficient must be simplified

A1

2

(ii)

$$\int 2x^{\frac{3}{2}} dx = \frac{2}{\frac{5}{2}} x^{\frac{5}{2}} \{+c\} \quad \{= \frac{4}{5} x^{\frac{5}{2}} \{+c\}\}$$

Can award for unsimplified form

B1

$$\int_0^4 8x^{\frac{1}{2}} dx - \int_0^4 2x^{\frac{3}{2}} dx$$

Ignore limits here

M1

$$= \frac{16}{3} (4)^{\frac{3}{2}} - 0 - \left[\frac{4}{5} (4)^{\frac{5}{2}} - 0 \right]$$

F(4) – F(0) used in either; {F(0) = 0 PI}
Cand. must be using F(x) as a result of
his/her integration in (b)(i) or in the (b)(ii)
B1 line above

M1

$$= \frac{256}{15}$$

Accept any value from 17.04 to 17.1
inclusive in place of 256/15

A1

4

(c) Translation

Accept 'translat...' as equivalent
[T or Tr is NOT sufficient]

B1

$$\begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided
linked to 'translation/move/shift'
(B0B0 if >1 transformation)

B1

2

[14]

Q34.

(a) (i) $p(-2) = -8 + 2 + 6$

$p(-2)$ attempted : **NOT** long division

M1

$$p(-2) = 0 \Rightarrow x + 2 \text{ is factor}$$

Shown = 0 plus statement

May make statement first such as " $x + 2$ is a factor if $p(-2) = 0$ "

A1

Minimum statement required "factor"

2

(ii) $b = -2$

No working required for B1 + B1

B1

$$c = 3$$

Try to mark first using B marks

B1

or long division/comparing coefficients

Award M1 if B0 earned and a clear method is used

(M1)

$$p(x) = (x + 2)(x^2 - 2x + 3)$$

*Must write final answer in this form if
long division has been used to get A1*

(A1)

2

(iii) $b^2 - 4ac = (-2)^2 - 4 \times 3$

*Discriminant correct from their quadratic
M0 if $b = -1$, $c = 6$ used (using cubic eqn)*

M1

$$b^2 - 4ac = -8 \text{ (or } < 0) \Rightarrow \text{no (other) real roots}$$

CSO All values must be correct plus statement

A1

Or $(x - 1)^2 + 2$

Completion of square for their quadratic

(M1)

$$(x - 1)^2 + 2 > 0 \text{ therefore no real roots}$$

Shown to be positive plus statement regarding no real roots

(A1)

Or $(x - 1)^2 = -2$ has no real roots

2

(b) (i) $(y_B =) 6$

Condone (0, 6)

B1

1

(ii) $\frac{x^4}{4} - \frac{x^2}{2} + 6x$

One term correct

M1

Another term correct

A1

All correct (ignore + c or limits)

A1

$$\left[\right]_{-2}^0 = 0 - (4 - 2 - 12)$$

$$= 10$$

F(-2) attempted

m1

$$= 10$$

CSO Clearly from $F(0) - F(-2)$

A1

5

(iii) Area of $\Delta = \frac{1}{2} \times 2 \times 6$

Condone - 2 and ft their y_B value

Or $\int_{-2}^0 (3x + 6) dx$ and attempt to integrate

M1

$$= 6$$

Must be positive allow -6 converted to +6

A1

Shaded region area = $10 - 6 = 4$

CSO 10 must come from correct working

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A1

3

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[15]

Q35.

(a) $(n =) -4$

Accept x^{-4}

B1

1

(b) $\left(1 + \frac{3}{x^2}\right)^2 = 1 + \frac{6}{x^2} + \frac{9}{x^4}$

Apply ISW after B2 stage
(B1 if correct but unsimplified seen)

B2,1,0

2

(c) $\int \left(1 + \frac{3}{x^2}\right)^2 dx = x - 6x^{-1} - 3x^{-3} + c$

At least one power of x correctly obtained in the integration of an expansion

M1

*A2 terms correct **and** “+ c”*

(A1F two terms in x correct ft on expansion provided integrating x to a negative power)

A2, 1, 0

3

(d) $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx = \left[x - \frac{6}{x} - \frac{3}{x^3}\right]_1^3$
 $= \left(3 - \frac{6}{3} - \frac{3}{27}\right) - \left(1 - 6 - 3\right)$

*Dealing correctly with limits; $F(3) - F(1)$
 (must have attempted integration to get F)*

M1

$= 8\frac{8}{9}$

CSO;

$\frac{80}{9}, \frac{240}{27};$

*OE provided value is **exact**, eg*

ISW dec value after exact value seen

NMS scores 0/2

A1

2

[8]